



Form1
Capstone Design Proposal
Design and Development of a WhatsApp-
Based Issue Reporting and Analytics
Dashboard System for TRAMOS Web at PT
Andakara Informatika Nusantara

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Submitted for
Capstone Design Project
to Faculty of Computer Science
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STATEMENT OF ORIGINALITY

In my capacity as an active student at President University and as the author of the Capstone Design Project stated below:

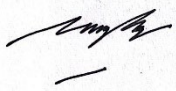
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I hereby declare that my Capstone Design Project entitled “**Design and Development of a WhatsApp-Based Issue Reporting and Analytics Dashboard System for TRAMOS Web at PT Andakara Informatika Nusantara**” is to the best of my knowledge and belief, an original piece of work based on sound academic principles. If there is any plagiarism detected in this final project, I am willing to be personally responsible for the consequences of these acts of plagiarism and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, 23 January 2026

Signer 1	Signer 2
	
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Design and Development of a WhatsApp-Based Issue Reporting and Analytics Dashboard System for TRAMOS Web at PT Andakara Informatika Nusantara

Approved:



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A. PROBLEM FORMULATION

1. Background of the Problem

At PT Andakara Informatika Nusantara, the Tracking Monitor Safety (TRAMOS) system helps manage trucks movement and supports logistics effectively. Although the platform has been widely adopted and used in real operational environment, the issue reporting process still manually. Instead of using built-in tools, users such as drivers and operators send reports through manual chats to just one person. If that person is away, nothing moves forward. Messages pile up without clear tracking, making it easy for requests to vanish unnoticed. Over time and delays grow simply because the workflow depends entirely on one link in the chain. Missteps in communication tend to reduce how well things work. When operations run fast, like in public transport, even small waits in information flow cause longer holdups down the line.

Another concern in this current process is the lack of systematic documentation. When problems arrive via manual chat, notes tend to be sketchy or absent altogether. That gap raises chances of lost details, mismatched files, and weak tracking paths. Tracking past incidents grows harder when data lacks shape. Judging how well fixes work feels guesswork without clear logs. Responsibility slips through cracks if nothing is formally noted down.

Without clear numbers on recurring issues, management struggles to spot patterns or measure effort spent resolving them. Week after week or month after month, similar problems reappear, yet tracking shifts over time remains impossible. Insights from earlier cases are missing, so responses stay reactive rather than preventive. When past outcomes aren't stored or reviewed, the same setbacks repeat unchecked. Sudden fleet interruptions happen more often when warnings go unseen. Data-backed tools such as TRAMOS Web become essential under these conditions.

2. Problem from a Customer Perspective

As a driver using the TRAMOS system, I am dissatisfied with the reporting process because every complaint must be funneled through manual chats to a single person. There are instance when such a person is not accessible or when there is a huge number of reports like when drivers are busy making deliveries, and as a result, there is a delay in replies.

As an operator, having a lack of documentation makes the problem worse for me because without documentation, each information on the incident reporting become disorganized in my head, so I end up listing the same incidents multiple times. This is not only wastes time for me but also makes me question the dependability of the system.

From our user perspective, not being able to generate statistical reports makes it difficult to receive an overview of general trends, like what type of issues are presented the most or how these issues tend to develop from one week to another. Furthermore, not being able to receive any type of foreknowledge about the issues makes it difficult to handle these issues proactively.

Taken as a whole, these issues create an inefficient and antiquated user experience in a industry where seamless support is crucial.

3. Problem Statement

The main problem in PT Andakara Informatika Nusantara's TRAMOS Web application is in its manual and central-based issue reporting and tracking solutions. Instead of automated systems, user complaints from drivers or operators go directly via chat to one person. Because of that single point of contact, response times slow down when message volume rises. When someone is unavailable, everything stalls. Information often slips through because entries aren't standardized. Without consistent logging, details get lost or duplicated, making it hard to follow up later. Management lacks clear data on recurring faults or peak trouble periods due to missing analytics tools. As a result, decisions happen without full context. Fixing root causes stays difficult under such conditions. Still, without prediction features, today's setup reacts after problems arise instead of foreseeing repeated glitches or offering early fixes drawn from past records. Hidden within everyday transport systems, this gap persists - unfixed errors disrupt vehicle performance and daily tasks, revealing how crucial it is to build smarter tools that streamline assistance and improve choices automatically.

4. Objective

The objectives of this project are to provide solutions for critical challenges in reporting issues and tracking supports TRAMOS Web on the PT Andakara Informatika Nusantara platform by developing an automated solution with scalability to improve efficiency:

- **Issue Reporting Automation:** Develop and create an AI chatbot for easy reporting of issues by users such as drivers and operators with automated responses, minimizing reliance on manual conversations with only one person.
- **Organized Ticket Management:** It is imperative to come up with an organized ticket management system to that is able to document and store reports from the users and issue logs in an organized manner that can easily be retrieved.
- **Analytics Dashboard:** Develop an analytics dashboard that provides managers and stakeholders with real-time data and statistics regarding trends, frequencies, types, and resolutions of issues.
- **Predictive Capabilities Integration:** The integration of machine learning capabilities for predictive analysis and recommendations related to potential problems in the future, moving from reactive to proactive assistance based on patterns of issues and anomalies in historical data.
- **Total Efficiency Enhancement:** Improving response time and customer satisfaction by optimizing support operations with technology, thus improving the efficiency of scalability and reliability of operations in logistics and trucks management. Through this, the project proposes to change the support structure on the platform by encouraging good decision-making and an environment that is responsive to users.

B. PROBLEM ANALYSIS

From the problem statements defined above, there are a number of key aspects to analyze for understanding the constraints and challenges in the present issue reporting and support handling process in TRAMOS Web. These actually mirror some operational realities in logistics and fleet management systems, where responsiveness, documentation, and data visibility remain crucial.

Aspect 1: Operational Efficiency and Scalability

The system in place for reporting current issues is not scalable in its operations. The reports of all issues are channeled through a process of communication to a focal point in a person-to-person manner. The system could work well in its current formulation when there is a small number of reports, like in peak operation hours when numerous system disruptions may occur.

When considering the operation, there are inefficiencies within the system that hinder fast response times, potentially causing some issues left unresolved. When applied in logistics and transportation, any inefficiency in resolving system or vehicle-related concerns can directly affect operations or deliveries. Research work conducted in service operation management views that inefficient, centralized operation, as in the case of service concerns, does not scale as effectively as needed when service requirements escalate, hence inefficient efficiency and pressures on service operation employees.

Aspect 2: Data Documentation & Traceability

Another important point here is the lack of documentation in a structured manner. Because issues are being recorded informally in chat messages, the documentation of issues is entirely reliant on human effort. Consequently, the risk of incomplete documentation, duplicates, or lost documentation is significant. In the long run, it becomes difficult to track the history with issues, ensure activities are taken on issues, or assess whether issues are effectively addressed for resolution.

From an engineer and system manager's perspective, it is important to document information properly to ensure traceability and accountability of information. Without an organized logging approach, it would be difficult for an organization to ensure reliable operational information is collected properly and maintained. This issue of traceability would make it difficult to audit and examine the system properly for improvement purposes.

Aspect 3: Decision Support and Data Availability

The major setback created by the lack of statistical reporting is that it hampers managerial attempts to provide insights into operational problems. This is because, through aggregated data, it is not possible for managers to identify problems or assess their resource intensity, let alone track trends that define such problems across time, hence reliance on assumptions in making managerial decisions.

Today in modern operation systems, data visibility is an integral aspect of ensuring improved service delivery. Research in information system management highlights the importance of analytical reporting in ensuring effective prioritization and planning, as it is able to convert operational data into meaningful information. This is an aspect of improved data visibility in an organization.

Aspect 4: Limitations of Preventive & Predictive

Presently, issue handling in TRAMOS Web is completely reactive. Troubleshooting and resolving issues happen only after they have arisen and are not capable of foretelling future instances when issues are likely to happen in future operations.

From a system engineering perspective, the inability to identify patterns and anomalies restricts continuous improvement. In logistics operations, recurring technical or operational issues often indicate underlying systemic problems. If these trends are not scrutinized for analysis of past data, they remain invisible and it would become difficult to have any early warnings to counter these problems.

C. SOLUTION SELECTION & SCENARIOS

Alternative Solution 1: Web-Based Manual Ticketing System with Basic Reporting

In this alternative, TRAMOS Web is enhanced with a web-based issue reporting and ticketing system. Drivers, operators, or even staff report their complaints by utilizing an organized form provided on the TRAMOS website. For each issue reported, it is automatically recorded on a database with the title of the ticket, where support staff or even administrators can browse and close these tickets based on their status of completion.

This system also allows for basic descriptive reports, for instance, the number of tickets for a certain period or category. These basic reports are very helpful in that they create a simplified view for reported problems but lack deep analytics or forecasts. This solution is centered on improving documentation and minimizing the use of manual chat conversations while preserving a relatively straightforward system architecture.

Alternative Solution 2: Web-Based AI ChatBot with Ticketing System

This alternative proposes the implementation of the web-based chatbot, which is part of the TRAMOS Web platform. The functionality of the chatbot allows the user to report problems, get automated responses based on the predefined knowledge base, and create support tickets if the problems aren't solved immediately.

All conversations and tickets are stored in a digital form that support staff can refer to. This chatbot alleviates the need for routine interaction through humans. This basic analytics module provides a summarization report regarding the chatbot engagement and the number of tickets. Although this approach brings in automation as well as intelligent response handling, it still has a limitation in that users need web platform access in order to submit any concerns they may have.

Alternative Solution 3 :AI WhatsApp Chatbot Integrated with Analytics and Insights Dashboard

This solution includes the integration of a chatbot using artificial intelligence and a WhatsApp interface with an “Analytics and Insight Dashboard” that is linked to “TRAMOS Web.” Users will be able to log complaints or report a problem straight from their “WhatsApp” account, as it is a service that is presently used in operations on a regular basis.

All conversations, tickets, and issue attributes are automatically stored in a structured database. These data are then processed and visualized through an analytics dashboard, presenting statistical reports, trends, and issue distributions. Over time, the collected data can be further analyzed to identify recurring patterns and support predictive insights for preventive actions. This solution emphasizes accessibility, scalability, and data-driven decision support, while minimizing operational friction for users.

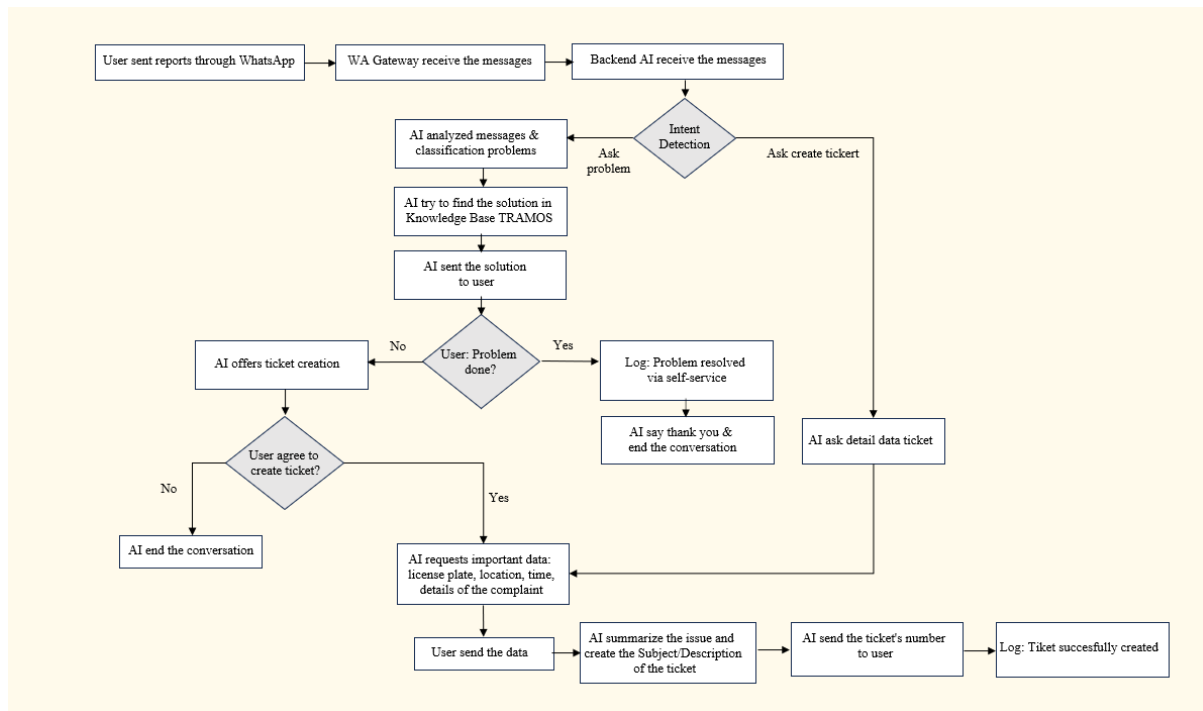
Solution Selection

On the basis of the analysis of the three alternative solutions put forward for resolution of the problem raised by the situation described, the best alternative solution is Alternative Solution 3 because it has the best level of accessibility and flexibility that can be utilized in logistical management involving the trucks where the users of the communication are highly mobile. They do not have time to respond using the web which is more time-consuming because it will open another web-related activity that disturbs their work activities. The use of WhatsApp makes it easy and convenient because it is already being used without putting extra effort that requires the use of the web. This also ensures that issues are expressed in real time without disrupting activities due to issues that are expressed online. The best part of this alternative solution is that it ensures all issues are linked to the central analytics dashboard that provides management with information on what is happening on the ground.

Solution	Advantages	Disadvantages
Alternative Solution 1: Web-Based Manual Ticketing	- Issues are documented systematically - Reduced dependency on individual staff - Simple and structured workflow	- Requires users to access the web - Limited real-time interaction - No automation or intelligent assistance - Reports are static and descriptive
Alternative Solution 2: Web-Based AI Chatbot	- Automated responses reduce manual workload - Structured issue submission - Improved consistency in data collection	- Still dependent on web access - Less practical for field users - Lower user adoption risk - Limited contextual accessibility
Alternative Solution 3: AI WhatsApp Chatbot + Analytics	- Highly accessible and user-friendly - Real-time interaction and scalability - Automated documentation and reporting - Enables trend analysis and predictive insights	- Requires WhatsApp Gateway integration - Initial setup and knowledge base training required

Solution Usage Scenarios

1. AI WhatsApp Chatbot Scenario

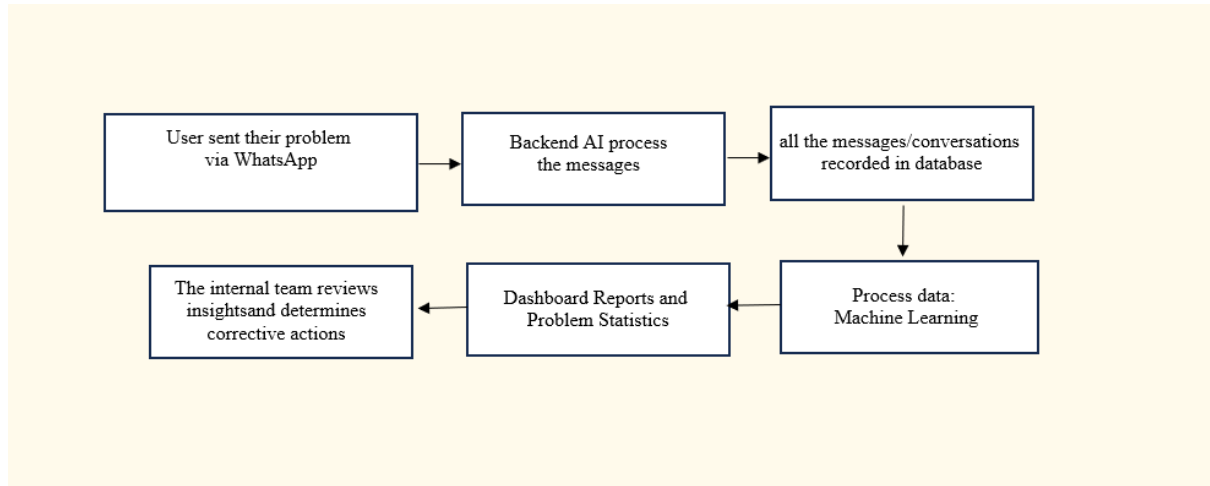


It is a diagram that explains the workflow of the AI-powered WhatsApp chatbot used to report any issue or generate a ticket in the TRAMOS Web system. The process begins with a message containing a description of the issue sent by a user over WhatsApp. The message will then be received and relayed to the backend AI system by the WhatsApp Gateway for intent identification and issue categorization based on which a solution to the issue is obtained from the TRAMOS troubleshooting knowledge base and sent back to the user over WhatsApp based on their intent.

If the user indicates that the problem is resolved, the conversation ends and is recorded as a self-service resolved case. This way, simple problems or recurring ones are solved without the need for manual processing. If there is no problem solution, it is suggested to generate a support ticket. When approved by the user, it requires data inputting, including trucks number, location, time, and description of the problem.

Once the required information is obtained, this AI system immediately produces a formal ticket through a summary of the problem and composing a suitable subject and description via natural language processing. The ticket system develops a new ticket and responds with a ticket number, subject, and closing message to this user through WhatsApp services. The process ensures quick response to issues and easy scalability of this support system.

2. Data Analytics Dashboard Scenario



This graph illustrates data flow from user interactions to analytics and insights. Each message, dialogue, and ticket information that is created from or through the use of the AI chatbot is recorded in a centralized database. These pieces of information are then processed by the backend application, allowing data analysis such as issue categorization and aggregation to be made. The processed data is used to generate visuals through an analytics dashboard that shows statistics about issues based on their resolutions.

The in-house team uses the dashboard to access insights and spot recurring trends or operations. Based on the insights, management decides on improvements to be made to system operations to ensure optimal system performance. This system allows the TRAMOS Web to translate raw system interactions into valuable insights that help in operations improvement decisions.

D. DEVELOPMENT EFFORT

1. Manpower Details

The development of this project involves two team members who play different roles and responsibilities to ensure effective collaboration and balance team efforts.

No	Name	Role and Responsibilities
1.	Cayla Winri Azzahra	Responsible for data-related components, including data processing, analytics dashboard development, and machine learning implementation. This role includes designing data pipelines, generating statistical reports, developing visualization for insights, and exploring predictive analysis based on historical issue data.
2.	Ivander Johana Pratama	Responsible for developing the AI chatbot system and WhatsApp Gateway integration. This includes handling message flow, intent detection, chatbot logic, ticket creation workflow, and backend integration between the chatbot and TRAMOS Web system.

2. Development tools

The following tools and technologies are used to support the development process:

- a. Visual Studio Code
Backend development, chatbot logic, and data processing.
- b. osTicket
Platform used as a Ticket Management System to handle issue tracking and ticket creation.
- c. Gemini AI
Used to support natural language understanding, response generation and automated interactions.
- d. Figma
Used for designing system flow diagrams and UI mockups, and analytics dashboard layouts.
- e. PostgreSQL (Docker-based)
The primary database system.
- f. Ngrok
Used during development and testing.
- g. Virtual Private Server (VPS)
Used to deploy backend services, chatbot processing, ticketing system, and analytics dashboard.
- h. XAMPP
Used to provide a local development for web services and backend testing.

3. Test Equipment

The testing process requires both hardware and software resources to ensure system reliability and functionality:

- Smartphone and Laptop
- Web Browser (Google Chrome)
- Internet Connection

4. Expert needs

No Expert needs for this project. All technical knowledge related to system development and data analysis can be handled by the project team.

5. Cost estimation

The project will ensure to minimize financial costs by utilizing open-source tools and existing development resources. However, there are certain components which will have a few operation costs.

Item	Estimated Cost	Description
WhatsApp Gateway Number	Low	A dedicated phone number is required for the AI chatbot to interact with users.
WhatsApp Gateway Service	Low	Cost related to message delivery and API usage (depending on provider).
VPS Hosting	Low to Moderate	Cost for hosting backend services, ticketing system, and analytics dashboard (if existing infrastructure is not available).
Gemini AI API Usage (if applicable)	Low	Potential cost depending on usage limits and API pricing tiers for natural language processing.

6. Timelines

The detailed timeline has been defined and is in line with the academic timeline provided by the faculty.

No	Task Title	Start Date	End Date	Duration (Days)	Progress	Timeline																															
						Des 25			Jan 26			Feb 26			Mar 26			Apr 26			May 26			June 26													
						W2	W4		W1	W4	W1	W4	W1	W4	W1	W5	W1	W4	W1	W4	W1	W4	W1	W4	W1	W4	W1	W4	W1	W4	W1	W4					
		8	9	26	5	6	29	30	2	3	25	26	27	2	3	27	30	31	1	2	29	30	1	4	5	28	29	1	2	26	29	30	1	2	3	6	7
1	Phase 1: FORMULATION					67%																															
1.1	Client Meeting & Identification Problem	12/8/25	1/7/26	23		100%																															
1.2	Scope Definition	1/8/26	1/12/26	3		100%																															
1.3	Proposal Writing & Review	1/13/26	1/23/26	9		0%																															
2	Phase 2: PLANNING					0%																															
2.1	Requirement Gathering (SOP Tramos)	1/26/26	2/6/26	10		0%																															
2.2	Functional Requirement	2/9/26	2/20/26	10		0%																															
2.3	Technology Stack Finalization	2/23/26	2/24/26	2		0%																															
2.4	Form-2 (Analysis) Writing & Review	2/25/26	2/27/26	3		0%																															
3	Phase 3: DESIGN					0%																															
3.1	System Architecture & Database Design	3/2/26	3/13/26	10		0%																															
3.2	Chatbot Flow & NLP Logic Design	3/16/26	3/20/26	5		0%																															
3.3	Dashboard UI/UX Wireframing	3/23/26	3/27/26	5		0%																															
3.4	Form-3 (Design) Writing & Review	3/30/26	4/1/26	3		0%																															
4	Phase 4: CONSTRUCT					0%																															
4.1	Module 1 Development: AI Chatbot	4/2/26	4/24/26	17		0%																															
4.1.1	WhatsApp API Setup & Integration	4/2/26	4/8/26	5		0%																															
4.1.2	NLP Training & Ticketing Logic	4/9/26	4/24/26	12		0%																															
4.2	Module 2 Development: Analytics	4/27/26	5/22/26	20		0%																															
4.2.1	Dashboard Frontend & Database Connect	4/27/26	5/8/26	10		0%																															
4.2.2	Machine Learning Model	5/11/26	5/22/26	10		0%																															
4.3	Integration & Form-4 Writing	5/25/26	5/29/26	5		0%																															
5	Phase 5: TEST					0%																															
5.1	System Testing (Internal Debugging)	6/1/26	6/5/26	5		0%																															
5.2	User Acceptance Test (UAT) with Client	6/8/26	6/12/26	5		0%																															
5.3	Feedback Revision & Bug Fixing	6/15/26	6/19/26	5		0%																															
5.4	Form-5 (Conclution) Writing & Review	6/22/26	6/26/26	5		0%																															
6	Phase 6: FINAL					0%																															
6.1	Deployment	6/29/26	7/3/26	5		0%																															
6.2	Final Report, Poster/Video Creation	7/6/26	7/17/26	10		0%																															

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